

1 Experiments: a small exact laboratory

Track experiments wrote a Python package, `bai` (`code/bai`), that computes exact posteriors over finite pools of theories made of DT° templates, reusing the code of AS (Claude (Anthropic), 2026a) for syntax, matching, Min and DTRC, and ran eight experiments, E1–E8. It is a laboratory for checking the claims of §§??–?? on small cases, not a prover. Three facts frame every number. Each posterior is exact *over its pool*, i.e. conditioned on “the theory is in the pool” (Remark 1.3). The derivation likelihood is a chain of at most two steps, so induction followed by MP and $\forall\text{E}$ is outside it, and the PA runs use L0. And E2, E3(b), E5(b), E8 sample multi-component theories at one fixed weight vector, where ?? does not cover the Dirichlet posterior; there the runs are a yardstick, not an instance of a theorem (??). Code lengths are in bits, $\alpha = \frac{1}{2}$, every run is seeded (Appendix A.5), and all findings below are computed unless a proposition is cited.

1.1 What is implemented

A *component* is a DT° template with a closed or open *guard* on each metavariable (may its body contain the parameter?); a theory is a finite set of components. The language is L_A ; no axiom of Q mentions $<$. There is one parameter p : a datum $\varphi(Sp)$ means $\forall y \varphi(Sy)$, and data with two parameters have probability 0. The instantiation grammar $\mathcal{Q}$ is a fixed PCFG over bodies (Appendix A.1) that satisfies ?? and counts holes and bound variables alike, so splitting a schema by the root of a body is an exact reparametrisation (??(b), ??). The prior is $\pi(T) \propto 2^{-\lambda \text{bits}(T)} (1 + |T|)^{-\tau}$, where $\text{bits}(T)$ is the length of T in a prefix code that satisfies Kraft’s inequality, $|T|$ is the template size and τ is the time exponent of ??; by default $\lambda = 1$ and $\tau = 0$. For instance $\{z + 0 = z\}$ costs 17.1 bits, $\{\forall x (x + 0 = x)\}$ 17.2 bits and $T^* := \text{Q1..Q7} + T_{\text{Ind}}$ 226.2 bits.

The mixture weights get a $\text{Dir}(\frac{1}{2})$ prior and are integrated out. L0 is citation: $P(s \mid T, w) = \sum_i w_i a_i(s)$ with $a_i(s) = \prod_M \mathcal{Q}(\theta_s(M))$ for the unique matcher θ_s . L1 is the forward chain Ch_J (??; rules in ??): cite component i with probability w_i , then take at most $J \in \{1, 2\}$ steps of $\forall\text{E}$ (term from $\mathcal{Q}_e$), Gen on p , or MP with a *cited* major premise, stopping with probability $c_{\text{stop}} = \frac{1}{2}$ while a rule applies; $c_{\text{ch}} := 1 - c_{\text{stop}}$. The chain never fails, so it needs no normaliser, and its law is linear in w . L1^{sel}_{cit} observes the chain through “closed quantifier-free sentences only” by per-citation rejection, a_i/c_i with c_i the probability that a chain from component i ends there (??). The marginal $P_T^{\text{Dir}}(D_n)$ is computed exactly by a dynamic programme over count vectors (??). Above 20000 states rigorous bounds replace it; the largest posterior mass a theory so bounded could have had never exceeded $8 \cdot 10^{-14}$ (Appendix A.4).

The pool at sample size n holds: hand-picked theories; theories built *causally* from D_b at build points $b \leq n$, namely Min of random subsets of D_b (AS Thm 3.3), skeleton clusters (DTRC without refutation, stopped at k clusters; AS §7.1) and, in PA runs, DTRC with the PA refuter and the observed-root split; in PA runs, $\text{Trim}(T, D_n)$ for every pool theory T , the components that could have produced a seen datum, so that $\text{Trim}(T^*, D_n) = \text{SeenQ}(D_n)$ is T_{Ind} plus the Q axioms cited so far; and $\text{Mem}(D_n)$. So every member uses only data already seen, which the track’s first version did not ensure (Remark 1.4). $T \vdash_J s$ iff some chain of at most J steps from T , with all rules enabled, ends in s : a sound under-approximation of derivability, Th_d of ?? with d in chain steps. $V_{\delta, J}$, the verifier $V_{\delta, d}$ of ?? on the pool, accepts s iff the pool posterior of $\{T : T \vdash_J s\}$ is at least $1 - \delta$; $J = 1$ in E1, E3(a), E4 and $J = 0$ in E2, E3(b), E7.

Proposition 1.1 (The computed likelihoods are exact). [*proved; computed*]

experiments Props X1–X4 (a) For a component τ and a sentence s with at most the parameter p , citing

data	lik.	mass of the generating theory at $n = 4, 8, 32, 256$	MAP at 256	$P(\vdash_1 \forall x\varphi)$	first n
<i>sch</i>	L0	0.87, 0.97, 1.000, 1.000	H_{sch}	$2 \cdot 10^{-7}$	16
<i>sch</i>	L1	0.84, 0.96, 1.000, 1.000	H_{sch}	$3 \cdot 10^{-7}$	16
<i>sch</i>	L1 ^{sel} _{cit}	0.43, 0.42, 0.53, 0.59	H_{sch}	0.484; 0.269	never
<i>all</i>	L1	0.87, 0.99, 1.000, 1.000	$H_{\forall}$	1.000	16
<i>allq</i>	L1	0.73, 0.90, 1.000, 1.000	$H_{\forall}$	1.000	16
<i>open</i>	L1	0.89, 1.000, 1.000, 1.000	H_{open}	1.000	8
<i>allq2</i>	L1	0.88, 0.997, 1.000, 1.000	$H_{\forall}$	1.000	16

Table 1: E1. Means over 5 seeds and, where the three formulas agree, over φ . $P(\vdash_1 \forall x\varphi)$ at $n = 256$; under L1^{sel}_{cit} it is 0.484 for $x + 0 = x$ and $0 + x = x$ and 0.269 for $\neg Sx = 0$. Last column: first n from which the generator keeps mass ≥ 0.99 , worst run. Under L0 (rows omitted) $H_{\forall}$ dies at its first instance; on *all* data H_{both} has mass ≥ 0.995 from $n = 32$, on *allq* data $H_{\forall} \cup H_{\text{open}}$ has 1.000 from $n = 16$, and on *open* and *allq2* data $\text{Mem}(D_n)$ has 1.000 from $n = 16$ and 64; $P(\vdash_1 \forall x\varphi) = 1.000$ at $n = 256$ in all four.

τ with independent bodies from $\mathcal{Q}$ yields s with probability $\prod_M \mathcal{Q}(\theta_s(M))$ if τ matches s by a matcher θ_s that respects the guards, and 0 otherwise. (b) For $a_{ij} \geq 0$, $\mathbb{E}_{w \sim \text{Dir}(\alpha)} \prod_{j=1}^n \sum_{i=1}^K w_i a_{ij} = \sum_z \prod_j a_{z_j j} \cdot \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_i \frac{\Gamma(\alpha+n_i(z))}{\Gamma(\alpha)}$, summed over assignments z of the data to the K components, with counts $n_i(z)$. The dynamic programme computes this sum, and the bounds bracket it. (c) For every theory, weight vector and J , the law of the chain’s output is a probability on sentences. (d) The backward recursion over predecessors computes the L1 coefficients $a_i(s)$ exactly whenever the predecessor sets it enumerates are complete. They are complete for the component shapes of Appendix A.2, and the code admits into L1 pools only theories of the shapes it checks (this left out 65 and 43 data-derived theories in two E1 configurations, none elsewhere).

Proof idea. (a) is the matching theorem, AS Thm 2.5. (b) expands the product and takes Dirichlet moments (??); the upper bound uses concavity of the log-likelihood on the simplex. (c) Every step chooses among finitely many outcomes of total probability 1, and no run fails. (d) Each predecessor set is in bijection with the matchers of a derived DT^o template, which are unique. Details in Appendix A.2. $\square$

The code agrees with brute force on 500 random Dirichlet sums ($1.4 \cdot 10^{-14}$) and with the referee’s independent reimplementations, written from the notes, on 400 marginals ($1.4 \cdot 10^{-14}$) and 904 L1 coefficients at $J = 1, 2$ ($2.1 \cdot 10^{-14}$); a forward sampler passes χ^2 tests; 21 unit tests pass.

1.2 Results

E1: $\forall x\varphi$ from data about φ . For $\varphi \in \{x + 0 = x, 0 + x = x, \neg Sx = 0\}$, seeds 0–4 and $n = 1, 2, 4, \dots, 256$, five generators are used: *sch* (L0 citations of $\sigma_{\varphi} = \varphi(z)$, closed terms); *all* (Ch₁ from $H_{\forall}$, closed $\forall E$ terms, so about half of the data are $\forall x\varphi$ itself); *allq* (the same with open terms, e.g. $\varphi(Sp)$); *open* (Ch₁ from H_{open} , whose Gen steps emit sentences such as $\forall y \varphi(Sy)$); and *allq2* (Ch₂ from $H_{\forall}$: $\forall E$ with an open term, then Gen). The pool (15–28 theories) holds $H_{\forall}$, H_{sch} , H_{open} , H_{both} , $H_{\forall} \cup H_{\text{open}}$, the over-specific $\{\varphi(0), \varphi(Sz)\}$, two root splits, three spare-slot theories, over-general generalisations of $\varphi(z)$ and F_0 , causal theories and $\text{Mem}(D_n)$. The likelihoods are L0, L1 with the generator’s own chain, and L1^{sel}_{cit} on *sch* data. Every generator has one component, so ?? applies (Table 1).

(1) *The factor of ??, exactly.* In Ch _{J} with closed $\forall E$ terms, $H_{\forall}$ emits $\forall x\varphi$ with probability c_{stop} and otherwise $\varphi(t)$ with probability $c_{\text{ch}} \mathcal{Q}(t)$, after which no rule applies; H_{sch} emits $\varphi(t)$ with

n	T^*	equiv.	weaker	unsound	acc.	MAP [tag] (count of 25)
8	$2 \cdot 10^{-21}$	$4 \cdot 10^{-13}$	0.744	0.256	2	skel4@8 [w] 12, DTRC@8 [w] 6, Trim(Q-lumped) [u] 4, ...
16	$9 \cdot 10^{-7}$	$5 \cdot 10^{-6}$	0.803	0.197	3	skel6@16 [w] 9, skel4@16 [w] 5, Q-lumped [u] 3, ...
32	0.197	0.197	0.727	0.077	0	$T^* - Q2$ [w] 7, skel6@32 [w] 6, T^* 5, ...
64	0.751	0.791	0.200	0.009	0	T^* 19, $T^* - Q3$ [e] 1, ...
128	0.991	0.991	0	0.009	0	T^* 25
512	0.996	0.996	0	0.004	0	T^* 25

Table 2: E2: posterior mass by tag (means over 25 seeds). “Acc.”: seeds in which $V_{0.05,0}$ accepts a false probe. Tags: [e] equivalent to T^* , [w] sound and strictly weaker, [u] unsound. At $n = 512$ the unsound mass is the false spare $T^* + \{0 = S0\}$.

probability $\mathcal{Q}(t)$. So on closed instances the $\log_2$ odds $H_{\text{sch}} : H_{\forall}$ gain $\log_2(1/c_{\text{ch}}) = 1$ bit per datum under L1 and stay at the prior odds under $L1_{\text{cit}}^{\text{sel}}$ ($c_{\forall} = c_{\text{ch}}$, $c_{\text{sch}} = 1$); this is experiments Prop X6 (proved), the Ch_J case of the *unnormalised* row of ???. Computed: $n + 0.090$ bits ($x + 0 = x$, $0 + x = x$) and $n + 1.440$ ($\neg Sx = 0$) at every n and seed, within $1.1 \cdot 10^{-11}$ bits of n plus the prior difference from the referee’s code; the $L1_{\text{cit}}^{\text{sel}}$ tie holds to 10^{-13} . (2) *The ω -gap*. Under $L1_{\text{cit}}^{\text{sel}}$ $P(T \vdash \forall x \varphi \mid D_n)$ tends to the prior share, in this code, of the provers among the theories tied with H_{sch} (??(b), ??), while held-out closed instances get mass ≥ 0.999 from $n = 8$ in every run except *open* data under L0 (??). (3) *The size principle*. At $n = 8$ on *sch* data, $\text{Mem}(D_n)$ has mass $\leq 2.4 \cdot 10^{-12}$ and the over-general theories $\leq 7 \cdot 10^{-5}$, both falling exponentially (??). This is one memoriser; the memoriser class can decay more slowly (??). Under L1 $\text{Mem}(D_n)$ is not provably the best memoriser, since a ground $\forall$ -sentence can raise its likelihood (the referee found $\text{Mem}(D_n) \cup \{\forall x (x + 0 = x)\}$ 11–16 bits worse, the margin shrinking). (4) *Quantified data*. From a prover of $\forall x \varphi$, L1 finds the generator by $n = 16$ and $P(T \vdash \forall x \varphi \mid D_n) \rightarrow 1$ (??(a), ??(a)). Under L0 derived sentences can only be axioms, so theories that list them win, and with Gen outputs of unbounded variety L0 ends at $\text{Mem}(D_n)$.

E2: PA without labels. Data are L0 citations from $T^* = Q1..Q7 + T_{\text{Ind}}$ (numbering of §??) at fixed weights 0.05 per Q axiom and 0.65 for T_{Ind} , with one-hole motives from $\mathcal{Q}$; seeds 0–24; $n = 8, \dots, 512$; the likelihood is L0 with the same $\mathcal{Q}$. Besides the causal theories, $\text{Trim}(T, D_n)$ and $\text{Mem}(D_n)$, the pool holds T^* , $T^* - Q_i$, **frag-complete** (Q plus the nine root fragments T_f), **frag-atoms** ($Q + \{T_-, T_+\}$), the spares $T^* + T_{\wedge}$, $T^* + \{\forall x (0 + x = x)\}$ and $T^* + \{0 = S0\}$, **IndSwap** (Q plus a swapped, equivalent induction template), two over-general induction templates, **Q-lumped** = $\{\forall x \forall y P(x, y), \forall x P(x), T_{\text{Ind}}\}$ and F_0 . Tags come from ?????? and a budgeted refuter (“unsound” is certain). The weights are fixed, so the applicable soundness statement is ??, not ?? (Table 2; SeenQ and code lengths in Table 5).

(1) T^* wins once Q1, Q2, Q4–Q7 have each been cited (the last at datum 20–97, median 55; Q3 is redundant by ??); the mass on its equivalents is ≥ 0.95 from $n = 32, 64$ or 128 (5, 15, 5 seeds). Before that the sound but weaker $\text{SeenQ}(D_n)$ leads, since an uncited axiom costs its prior bits and the factor of ??(a). (2) At $n \leq 16$ **Q-lumped** or its trim beats SeenQ in some seeds (by up to 12.7 bits), and $V_{0.05,0}$ then accepts $\forall x (x + 0 = 0)$, $\forall x (Sx = x)$, $\forall x \forall y (y = x)$: in 5 of 25 seeds, never from $n = 32$ (19 of 25 with the first version’s pool, Table 5). This is consistent with ??, which needs $\delta_n \leq 2^{-226} R_{\alpha}(n, 8) \delta' \asymp 2^{-226} n^{-3.5}$ (??). (3) **frag-complete** has T^* ’s instance set and its laws include T^* ’s (??(b)), so it loses the prior difference (698.9 bits) plus 4.2 bits per doubling of n ($n = 64 \dots 512$; asymptotically $(16 - 8)/2 = 4$, the Occam term of ??(c)). (4) Unused spare sentences lose 0.49 bits per doubling and the nested $T_{\wedge}$ 0.25 (??(a), (c)); the inconsistent

data	MAP at the largest n (5 seeds; L0 and L1 alike)	inst. set	gain	$P(\vdash \text{ held-out})$	superset
<i>heavy</i>	C_2 (3 seeds), $\{\varphi(z), \varphi(SSz)\}$ (2)	same	$212 \rightarrow 455$	1.000	$3 \cdot 10^{-63}$
<i>numerals</i>	R_9 or R_{10} at $n = 512$; R_{11} or R_{12} at 1024	subset	$1271 \rightarrow 2897$	0.333	0
<i>small</i>	H_{sch} up to $n = 512$; $\text{Mem}(D_n)$ at 1024 (L0)	same $\rightarrow$ finite	$0 \rightarrow 156$	$1.000 \rightarrow 0.000$	$7 \cdot 10^{-5}$
<i>skewQ</i>	frag1, a root split (from $n = 512$)	same	$70 \rightarrow 212$	1.000	$2 \cdot 10^{-20}$

Table 3: E3(a). Gain: code length of H_{sch} minus that of the MAP under L0, bits, $n = 512 \rightarrow 1024$ (means). $P(\vdash \text{ held-out})$: mean mass deriving $\varphi(9)$, $\varphi(2 + 3)$, $\varphi(1 \cdot 4)$. Superset: largest mass on over-general theories at the largest n over seeds and likelihoods. Families: Table 6.

motives	MAP at $n = 256 / 1024 / 2048$ (5 seeds)	equiv.	weaker	T^*	unsound at 16
<i>dtrc-motive</i>	$T^* / T^* + T_{\exists} / T^* + T_{\exists}$	1.000	0	$5 \cdot 10^{-70}$	0.25; 1.0
<i>root-skew</i>	$T^* + T_{\rightarrow} / \text{frag-observed@16, @32} / \text{same}$	1.000	0	0	0.52; 1.0
<i>deep</i>	$T^* / T^* / T^*$	0.998	0	0.998	$2 \cdot 10^{-11}$; $7 \cdot 10^{-11}$
<i>atomic</i>	frag-atoms (4), T^* (1) / frag-atoms / frag-atoms	$2 \cdot 10^{-178}$	1.000	$4 \cdot 10^{-298}$	0.21; 0.999

Table 4: E3(b). Masses at $n = 2048$; unsound mass at $n = 16$ as mean; max over seeds. Code lengths relative to T^* at $n = 2048$: *root-skew* frag-observed@ b -1387 to -1509 bits; *dtrc-motive* $T^* + T_{\exists}$ -228 to -318 ; *deep* frag-complete $+665$ to $+674$; *atomic* frag-atoms -986 to -1028 .

$T^* + \{0 = S0\}$ keeps 0.004 at $n = 512$. (5) The equivalent but uncited **IndSwap** has likelihood 0 from the first induction datum: citation identifies the cited axioms. Over-general induction templates lose linearly.

E3: misspecified usage. All data are theorems of the target, but their law is no pool theory’s (??). (a) Instances of $\varphi = (x + 0 = x)$ under other term laws: *heavy* (a zeta(1.5) numeral with probability 0.7, else $S^k(t_1 \circ t_2)$), *numerals* ($S^k 0$, $k \sim \text{Geom}(0.2)$), *small* (Q -terms with at most 4 symbols: 12 distinct sentences) and *skewQ* (another PCFG); L0 ($n \leq 1024$) and L1 (Ch_1 , $n \leq 512$); seeds 0–4. The pool adds to E1’s hand pool the closed-guard numeral splits $R_m = \{\varphi(0), \dots, \varphi(S^{m-1}0), \varphi(S^m z)\}$, $m \leq 16$, and the nested $C_1 = \{\varphi(z), \varphi(Sz)\}$ and $C_2 = C_1 \cup \{\varphi(SSz)\}$. (b) E2’s T^* and weights with other motive laws: *dtrc-motive* (AS’s motive generator), *root-skew* (roots $\rightarrow, \wedge, =, \exists$ with .5, .2, .2, .1), *deep* (more connectives) and *atomic* ($a(x) = b(x)$ with probability 0.6, $a(x) < b(x)$ with 0.4); L0; $n \leq 2048$; E2’s pool plus $T^* + T_f$ for every root f (Tables 3 and 4).

(a) The posterior moves at a linear rate to the best-fitting member of a family; the winner keeps changing as n grows. On numerals the best split deepens (R_4/R_5 at $n = 128$, R_9/R_{10} at 512, R_{11}/R_{12} at 1024, interior to the family), like the drift of ??(d) in track universal’s calculus; $n = 1024$ is too small to test its part (b). On heavy tails the deepest nesting available wins, and a deeper one would probably win if the exact programme could hold it (conjecture). Nothing unsound survives at large n , but completeness is lost: with numerals only the held-out $\varphi(2 + 3)$, $\varphi(1 \cdot 4)$ are not derived, and with a finite selected support the axioms become the 12 observed sentences. As in ??, all data are theorems of the target and the posterior goes to a strictly weaker theory. L1 gives the same MAPs as L0 for $n \leq 512$. (b) Under connective-rich motive laws the winners contain T_{ind} or a connective fragment and Q1, Q2, Q4–Q7, so they are equivalent to T^* (??); yet under *dtrc-motive* and *root-skew* the MAP leaves T^* : the theorems survive, the axioms

do not. Under *atomic* motives T^* has mass 0.988 at $n = 64$ in 4 of 5 seeds, and then $Q + \{T_=: T_<\}$ takes mass 1.000 in every seed from $n = 1024$. That theory does not prove the irrationality of $\sqrt{2}$, which T^* proves (??): the posterior moves *away* from T^* to a strictly weaker theory (??). Unsound lumps recur at $n = 16$.

E4: an adaptive prover. The prover wins if $V_{\delta,1}$ ever accepts some s with $T^* \not\vdash_1 s$. T^* always has one component, so ?? bounds the win probability by δ' when $\delta = w^*\delta'$ (Table 7). (A) The tight construction of IL Prop C.5 (Claude (Anthropic), 2026b) with a Dirichlet decoy: $T^* = \{\varphi(z)\}$ on numerals with $P(t=0) = 1 - u$, decoy $T' = \{\varphi(0), 0 = S0\}$, prior w^* on T^* ; the prover asks $0 = S0$ every round. It wins iff the first t^* data are $\varphi(0)$, with t^* the first n at which $\pi_n(T^*) \leq \delta$ given $\varphi(0)^n$, so $P(\text{win}) = (1 - u)^{t^*}$ exactly. (B, C) $\varphi = (x + 0 = x)$, $T^* = H_{\text{sch}}$, E1's hand pool plus $\text{Mem}(D_n)$; $w^* = 7 \cdot 10^{-6}$ (the pool version of ??), bound 0.05; each round the prover asks eight sentences that T^* does not derive; 100 streams, every $n \leq 256$, L0 and L1. Well specified, wins stay below δ' : 9–46 times below in (A), whose Dirichlet decoy is slower than IL's fixed-weight one (??), and 0/100 in (B), where the largest invalid mass (0.45, 0.53) sits on the true but underivable $\forall x (x + 0 = x)$ at $n \leq 6$. Misspecified, the prover wins every time when 10% of the data are false near misses $t + 0 = St$ and the mistake schema is in the pool (C1), and when only a decoy fits selected data better ((A) with only $\varphi(0)$ shown); heavy tails (C2) and selection with a clean memoriser in the pool (C3) did no harm. No theorem covers the misspecified case (??).

E5: spare slots, L_∞ against L_5 , and Gold's text (Table 8). (a) Exact Bayes factors of $\{\varphi(z)\}$ plus a spare against $\{\varphi(z)\}$ on its own L0 data: the unused spares ($0 = S0$; $z \cdot 0 = 0$) match ??(a) to all printed digits (slope -0.500 per doubling); the nested $\varphi(Sz)$ has slope -0.281 on $n = 256 \dots 4096$ and, by quadrature, -0.251 ± 0.002 on $10^2 \dots 10^8$, supporting the $n^{-1/4}$ of ??(c), itself a proof sketch. A false spare sentence is thus removed by positive data only like $n^{-1/2}$. (b) With L_k the k ground sentences $\varphi(S^j 0)$, $j < k$, and $L_\infty = \{\varphi(z)\}$ under Q_{num} : on data from L_5 the posterior switches to L_5 between $n = 64$ and 128 (mass 1.000 from 256, 10/10 seeds), and on data from L_∞ it keeps L_∞ (1.000 from $n = 16$). ??(a) proves this for fixed generators; here L_5 has Dirichlet weights and is sampled at one weight vector, so the run is a yardstick (??). (c) Against the MAP learner over $\{L_1, \dots, L_{60}, L_\infty\}$ the first 14 stages of Gold's construction all end (740 data): the start of a text on which, if every stage ends, the MAP changes infinitely often (??(b), (c)).

E6: generating axiomatisations. Eight forms of commutativity of $+$ are compared.

Proposition 1.2 (Forms of commutativity). *[proved; computed]*

experiments Prop X12 Let $A_{xy} := \{\forall x \forall y (x + y = y + x)\}$, $A_{yx} := \{\forall y \forall x (x + y = y + x)\}$, $S_{ab} := \{z + z' = z' + z\}$, $M_x := \{\forall x (x + z = z + x)\}$ and $M_y := \{\forall y (z + y = y + z)\}$, with closed guards, and let S_{ab}^{open} , M_x^{open} , M_y^{open} be the same templates with open guards. (a) A_{xy} , A_{yx} , M_x^{open} and M_y^{open} are logically equivalent. (b) $S_{ab} \cup M_x \cup M_y \cup S_{ab}^{\text{open}} \not\vdash \forall x \forall y (x + y = y + x)$. So S_{ab} , M_x , M_y and S_{ab}^{open} are strictly weaker than A_{xy} , although S_{ab} , M_x and M_y yield exactly its closed instances $t + t' = t' + t$.

The proof (Appendix A.3) uses the structure $\mathbb{N} \sqcup \{\alpha, \beta\}$ with $\alpha + \beta = \alpha \neq \beta = \beta + \alpha$. Part (a) corrects a remark of the track's referee that open guards cannot reach $\forall x \forall y$ under the single-parameter convention: one bound variable and the parameter already give two variables. The

fixed pool is the eight forms, $A_{xy} \cup A_{yx}$, $A_{xy} \cup S_{ab}$ and $\text{Mem}(D_n)$. Data come from **Ch₂** (closed $\forall$ E terms) on A_{xy} and on M_x , and from **L0** citations of S_{ab} ; seeds 0–4; $n \leq 256$ (Table 9).

L1 identifies the generating form by $n = 32$ in every run, among logically equivalent and among instance-equivalent weaker forms (compare ??); A_{xy} is told from A_{yx} and the open forms by partial instances such as $\forall y (2 + y = y + 2)$. Under **L1_{cit}^{sel}** the five closed-guard forms have equal likelihoods on every datum (as in E1(1)), so the mass on provers of commutativity tends to their prior share 0.088, not to 1: the ω -gap of ?? again, not a tie among equivalents. A redundant second form is a spare slot ($< 10^{-9}$; ??).

E7: time factor and steeper penalty (E2’s data and pools, 25 seeds; Table 11). The time factor costs $\tau \log_2(1 + |T|)$ bits (T^* has 77 symbols), so at $\tau = 1$ it moves the log-odds between T^* and any hand competitor by at most $\log_2(78/2) \approx 5.3$ bits. It changes nothing visible from $n = 64$ on; at $n \leq 32$ it slightly favours small unsound lumps (unsound MAP at $n = 8$ in 9 instead of 7 seeds at $\tau = 1$; largest change of a reported mass 0.181 at $\tau = 1$, 0.563 at $\tau = 4$). This follows from the factor’s definition and does not test the brief’s H6 (????). The steepness $\lambda = 2$ makes early unsound lumping far more frequent (unsound MAP at $n = 8$ in 22 seeds) but removes the false spare faster; $\lambda = \frac{1}{2}$ does the reverse (??).

E8: splits under a derivation likelihood. Let $\varphi(t) := (t+0 = t)$, $\psi(t) := (0+t = t)$, $\text{whole} := \{\varphi(z), \psi(z), \varphi(z) \rightarrow \psi(z)\}$, and $\text{split} := \text{whole}$ with $\varphi(z)$ replaced by $\varphi(0), \varphi(Sz), \varphi(z+z'), \varphi(z \cdot z')$ ($K = 4$ root productions). Under **Ch₁** a citation of $\varphi(t)$ stops or applies MP with the cited conditional, giving $\psi(t)$, which can also be cited: the component behind a ψ -datum is latent. Data come from **Ch₁** on whole at fixed weights (0.5, 0.25, 0.25), with bodies from $\mathcal{Q}$ (*wellspec*) or from another PCFG (*skew*: roots 0, S , $+$, $\cdot$ with .5, .1, .2, .2); seeds 0–9; $n \leq 4096$. Table 10 gives Bayes factors without the prior, which adds 76.4 bits for whole.

At matched weights ($w_{\varphi p_f}$) the two theories give every datum the same **L1** probability, to $1.1 \cdot 10^{-13}$ (experiments Prop X13, ??). With learned weights and well-specified usage the split loses 1.43 bits per doubling under **L1** and 1.51 under **L0** ($n = 256 \dots 4096$; 1.60, 1.76 on 1024 $\dots$ 4096), near the $(K - 1)/2 = 1.5$ of conjecture ??(d). With misspecified usage it wins linearly, more under **L1** (0.202 against 0.074 bits per datum), because there the term of a ψ -datum also informs the split (??(c)). E8 does not test ??(b), which concerns a one-component schema with fixed weights. In this model a derivation likelihood does not change the MDL finding (??); what it changes in PA, the consequences of a split (??), the chain cannot test.

1.3 What the experiments do not show

Remark 1.3 (Limitations). [(i) computed; (ii)–(v) proved (properties of the code)]^{experiments §§11, 12, 16.5}
(i) *Pool, not class.* The winners, and even the frequency of unsound acceptance, changed with the pool: E2’s acceptances fell from 19 to 5 of 25 seeds with causal pools and trimmed theories, and E3(a)’s winners moved when the families R_m , C_k were extended. The pool version of ?? keeps soundness under pool restriction; nothing comparable is shown for identification. Search beyond a pool is open. (ii) *Chains, $J \leq 2$.* MP needs a cited major premise, so no run computes PA under a derivation likelihood. (iii) *One parameter; fixed $\mathcal{Q}$.* A misfit of usage statistics is absorbed by extra components (E3, E8); a learned grammar can change the sign of the Occam drift (??). (iv) *Bounded derivability.* $\vdash_J$ under-approximates provability, so the $P(T \vdash s)$ columns are lower bounds. (v) *Small runs.* 25 seeds in E2 and E7, 10 in E5 and E8, 5 otherwise; $n \leq 4096$ (quadrature to 10^8). The exact Dirichlet sum is exponential in the number of overlapping

components, which also stops E3(a)’s nesting at C_2 . No run uses human-written theorem lists: every misspecified generator perturbs or selects from the model’s own term or motive law. Neither Hänni’s collapse (??) nor his “do not contradict” variant is implemented.

Remark 1.4 (Claims of the first version). [refuted]

experiments §16.4; referee M1–M4 Each claim is kept with its counterexample. (1) E2: “in 3 of 5 seeds an unsound lump has mass ≥ 0.997 at some $n \leq 32$ and a $\delta = 0.05$ verifier accepts a false sentence”. True of a pool that used unseen data and lacked $\text{SeenQ}(D_n)$; with causal pools, 2 of seeds 0–4 and 5 of 25. Counterexample to its representativeness: seed 2 at $n = 32$, where the old `skel4` (built from 40 data) had mass 1.0 and the causal `skel4@32` has 0.85. (2) E2: “MAP at $n = 8$: $\text{DTRC}(n = 16)$ three times”: built from data 9–16, unseen at $n = 8$. (3) E6: “the five closed-guard forms are logically equivalent”: Proposition 1.2(b). (4) “Soundness at a constant threshold for any well-specified data”: fails for multi-component T^* at fixed weights (??). (5) PA: “robust at the level of theorems, not of axioms”: E3(b) atomic motives (??). (6) E1: “ $P(T \vdash \varphi(t^*) \mid D_n) = 1.000$ from $n = 8$ in every configuration”: *open* data under L0 give 0 at $n = 4, 8$. (7) E2: “no false probe is accepted”: the mean over seeds hid seeds 1, 2 and 4.

A Experiments: proofs and details

A.1 Grammar, prior code and pools

Q. A term node is 0, S , $+$, $\cdot$ with weights .45, .35, .10, .10, a variable with total weight .3 split uniformly over the variables in scope (holes and bound variables alike), or p with weight .15 where the guard allows it. A formula node is $=$, $<$, $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\forall$, $\exists$ with weights .35, .20, .10, .10, .05, .10, .02, .05, .03. Weights are renormalised over the options available. The mean number of same-sort children is 0.75 for closed terms, at most 0.75 for other terms and 0.72 for formulas, so bodies are finite almost surely; the referee’s exact size distributions agree.

Prior code. $\text{bits}(\tau)$ is $-\log_2$ of a subcritical stochastic grammar over templates (formula and term nodes, metavariable occurrences, arities and guards); a theory of m distinct components costs $|\gamma(m)| + \sum_i \text{bits}(\tau_i) - \log_2 m!$ with γ the Elias code. The referee’s independent implementation gives the same lengths to four decimals; Kraft’s inequality holds (??).

Pools. Theories are deduplicated at each n by their components (canonical template plus guards); if $\text{Mem}(D_n)$ equals a pool theory, that theory keeps its mass and Mem becomes an alias without mass (the first version counted it twice, which affected E6 at $n = 1, 2$ only). Under L0 $\text{Mem}(D_n)$ is the best memoriser covering D_n : a further ground sentence costs prior bits and the factor of ??(a) and cannot raise the likelihood. Build points: E1 $b \in \{1, 2, \dots, 64\}$ (powers of 2); E3(a) $\{8, 32, 64\}$; E2, E3(b), E7 $\{8, 16, 32, 64\}$; E4 and E6 have none. In E2 a theory of Q axioms and induction-type templates is tagged equivalent to T^* iff it contains Q1, Q2, Q4–Q7 and a full-induction template (T_{Ind} , `IndSwap` or a connective fragment); other tags come from a budgeted refuter, and an unrefuted theory is “unknown”. The false probes are $0 = S0$, $\forall x (x + 0 = 0)$, induction with a wrong base, $\forall x (Sx = x)$ and $\forall x \forall y (y = x)$; held-out induction instances are drawn outside the training stream.

$\text{L1}_{\text{cit}}^{\text{sel}}$. The class probabilities c_i are exact for components $\forall^a \chi$ with χ quantifier-free, term metavariables only, every leading variable used and no MP components (a finite Markov chain on the number of leading $\forall$ and the presence of p); other components are left out of $\text{L1}_{\text{cit}}^{\text{sel}}$ pools. Monte Carlo agrees on the eight E6 forms (40000 chains each; largest $|z| = 2.17$).

A.2 Proof of Proposition 1.1

(a) Bodies from $\mathcal{Q}$ contain no parameter other than p , so $\tau\theta$ is already in closure-normal form and the event “the datum is s ” is the event $\tau\theta = s$. By AS Thm 2.5 at most one θ satisfies $\tau\theta = s$, so $P(\tau\theta = s) = \sum_{\theta:\tau\theta=s} \prod_M \mathcal{Q}(\theta(M))$ has at most one term, $\prod_M \mathcal{Q}(\theta_s(M))$. A body containing p has probability 0 under a closed guard. $\square$

(b) Expanding, $\prod_j \sum_i w_i a_{ij} = \sum_z \prod_i w_i^{n_i(z)} \prod_j a_{zjj}$, and the Dirichlet moments $\mathbb{E} \prod_i w_i^{n_i} = \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_i \frac{\Gamma(\alpha+n_i)}{\Gamma(\alpha)}$ give the formula (??). *The programme.* The summand depends on z only through $\prod_j a_{zjj}$ and the counts. A datum covered by one component contributes a fixed factor and one count. For the others, with $S_j := \{i : a_{ij} > 0\}$, the sum over their assignments is the polynomial $\prod_j \sum_{i \in S_j} a_{ij} x_i$ evaluated by the linear functional $x^c \mapsto \prod_i \Gamma(\alpha + c_i + c_i^0) / \Gamma(\alpha)$ (c^0 the fixed counts), times $\Gamma(K\alpha) / \Gamma(K\alpha + n)$. A dense array (at most three overlapping components) or variable elimination over count vectors computes the coefficients exactly. In the latter a component’s factor is applied, and its count dropped, once no later group contains it; this is valid because the functional is a product over components. *The bounds.* All terms are nonnegative, so any single term is a lower bound. For the upper bound, $\mathbb{E}_w L(w) \leq \max_w L(w)$ with $L(w) = \prod_j \sum_i w_i a_{ij}$, and $f := \ln L$ is concave on the simplex. For an interior w with $L(w) > 0$ and any w° in the simplex, $f(w^\circ) \leq f(w) + \nabla f(w) \cdot (w^\circ - w) \leq f(w) + \max_i \partial_i f(w) - n$, since $\nabla f(w) \cdot w^\circ \leq \max_i \partial_i f(w)$ and $w \cdot \nabla f(w) = \sum_j \sum_i w_i a_{ij} / \sum_k w_k a_{kj} = n$. $\square$

(c) At most J rules are applied. Each step chooses among finitely many outcomes (stop, or a rule that applies) with probabilities summing to 1; each rule’s input (a term from $\mathcal{Q}_e$, bodies from $\mathcal{Q}$, a uniform choice among the applicable MP components) has a probability distribution ($\mathcal{Q}$ is proper). A rule is applied only when it applies, so no run fails; the runs have total probability 1, and the datum is a function of the run. $\square$

(d) Let $G_k(s)_i$ be the probability that the chain reaches step k at s after citing component i , divided by w_i ; so $G_0(s)_i$ is the L0 coefficient of (a). For $1 \leq k \leq J$ the chain leaves s' only if a rule applies to it, then takes rule r with probability $(1 - c_{\text{stop}})c_r / \sum_{r'} \text{applies to } s' c_{r'}$ and input ξ with probability $P(\xi)$. Hence

$$G_k(s) = \sum_{(s', r, \xi): r(s', \xi) = s} G_{k-1}(s') (1 - c_{\text{stop}}) \frac{c_r}{\sum_{r'} \text{applies to } s' c_{r'}} P(\xi), \quad a_i(s) = \sum_{k \leq J} G_k(s)_i \text{stop}_k(s),$$

with $\text{stop}_k(s) = 1$ if $k = J$ or no rule applies to s , and c_{stop} otherwise. The recursion is exact iff every predecessor s' with $G_{k-1}(s') > 0$ is enumerated. The following give complete predecessor sets.

- (d1) $\forall E$ *guided by templates.* Let $U = \forall x B$ be a component whose derived template $D := B[u/x]$ (u a fresh 0-ary term metavariable) is in DT° . Then x is no argument of a metavariable occurrence (else an argument would contain u), so x occurs only at rigid positions, bodies never contain x , and $(B\theta)[t/x] = D(\theta \cup \{u \mapsto t\})$. So the pairs (θ, t) with $\forall E(U\theta, t) = s$ correspond to the matchers of D against s , of which there is at most one (AS Thm 2.5). If x does not occur in B , the predecessor is $\forall x s$, with total weight $\sum_t \mathcal{Q}_e(t) = 1$.
- (d2) $\forall E$ *from P or $\forall x P(x)$ at step 0.* For a parameter-free s the predecessors are $\forall x \psi_S$, with ψ_S the sentence s with a nonempty set S of occurrences of one closed term t replaced by x . Since the PCFG probability is a product over nodes and every node of ψ_S lies in a context with a variable, $\mathcal{Q}(\forall x \psi_S) = \mathcal{Q}(\forall x s) \prod_{o \in S} r_o$ with $r_o := (p_{\text{var}}/\nu_o)/\mathcal{Q}_v(t)$; here p_{var} is the normalised variable weight, ν_o is 1 plus the binder depth at o , and $\mathcal{Q}_v(t)$ is the probability of t in a context with a variable. So $\sum_{S \neq \emptyset} \mathcal{Q}(\forall x \psi_S) = \mathcal{Q}(\forall x s) (\prod_o (1 + r_o) - 1)$; for $\forall x P(x)$ the same

holds up to the factor of the root $\forall$. The code uses this closed form only in theories without MP components, where these parameter-free predecessors admit only $\forall E$. If s contains p and the guard is closed, t contains p and S is all occurrences of t .

- (d3) *Gen, then $\forall E$ ($J = 2$).* A step-1 Gen output is $\text{gen}(s_0) = \forall x s_0[x/p]$ for a cited $s_0 = T\theta_0$ containing p , and $\forall E(\text{gen}(s_0), t) = s_0[t/p]$. Let D be the component with its rigid occurrences of p replaced by a fresh 0-ary metavariable u , and suppose $D \in \text{DT}^\circ$ (no rigid p inside an argument; the track's notes leave this hypothesis implicit). Then $s_0[t/p] = D\theta$ with $\theta(M) = \theta_0(M)[t/p]$ and $\theta(u) = t$, so θ is the unique matcher of D against s . Closed-guard bodies satisfy $\theta_0(M) = \theta(M)$; open-guard bodies $\theta_0(M)$ arise from $\theta(M)$ by replacing some occurrences of t by p ; without a rigid p , t occurs in some open-guard body. The code enumerates exactly these candidates and keeps those with $\forall E(\text{gen}(s_0), t) = s$.
- (d4) *Gen and MP.* $\text{gen}^{-1}(\forall x \chi) = \chi[p/x]$ if χ uses x and s has no parameter. For an MP component $A \rightarrow B$, $B \in \text{DT}^\circ$ contains A 's metavariables, so the matcher θ of B against s is unique and the predecessor is $A\theta$.
- (d5) *Pruning.* A candidate is skipped only if its ($\forall$ -spine, parameter flag) is outside the abstract set of step $k - 1$, computed forward from the components; by induction on k every step- k sentence lies in that set. $\square$

The code admits into L1 pools only theories whose $\forall$ -rooted components have derived templates in DT° (twice derived at $J = 2$) and whose formula-metavariable components are P or $\forall x P(x)$ at $J = 1$ (none at $J = 2$). A term with more than 12 occurrences in a datum stops the enumeration and raises a counter; it was 0 in every run. The support check does not test the hypothesis of (d3). It holds for every component without a rigid p , which includes every hand theory of E1 and E6; for data-derived theories at $J = 2$ (E1 *allq2*) exactness rests on the computed checks: the guided recursion equals a brute-force recursion over all subsets of all occurrences to 10^{-9} on samples from 10 theories at $J = 1, 2$ (same class, so not independent), and the referee's own recursion agrees on 904 triples at $J = 1, 2$ to $2.1 \cdot 10^{-14}$.

A.3 Proof of Proposition 1.2

(a) A_{xy} and A_{yx} differ by the order of two universal quantifiers. M_x^{open} cites $\forall x (x + p = p + x)$ (body $z := p$), which under the closure reading is $\forall w \forall x (x + w = w + x)$, i.e. A_{yx} ; and each of its instances $\forall x (x + b = b + x)$, b a term with at most the parameter p , follows from A_{xy} by $\forall E$ (and Gen on p under the closure reading). Likewise M_y^{open} cites $\forall y (p + y = y + p)$, i.e. A_{xy} .

(b) Take the domain $\mathbb{N} \sqcup \{\alpha, \beta\}$, standard on $\mathbb{N}$, with $<$ arbitrary, $S\alpha = \alpha$, $S\beta = \beta$, and for $a \in \{\alpha, \beta\}$, $n \in \mathbb{N}$: $a + n = n + a = a$; $a \cdot n = n \cdot a = a$ if $n \neq 0$ and 0 if $n = 0$; $\alpha + \alpha = \alpha \cdot \alpha = \alpha$, $\beta + \beta = \beta \cdot \beta = \beta$, $\alpha + \beta = \alpha \cdot \beta = \alpha$, $\beta + \alpha = \beta \cdot \alpha = \beta$. Each $\mathbb{N} \cup \{a\}$ is closed under S , $+$, $\cdot$ and $+$ is commutative on it. Closed terms denote naturals, so the closed instances of S_{ab} hold. An instance $\forall x (x + t = t + x)$ of M_x , t closed of value n , holds for $x \in \mathbb{N}$ by arithmetic and for $x = a$ since $a + n = a = n + a$; likewise for M_y . An instance $\forall w (f(w) + g(w) = g(w) + f(w))$ of S_{ab}^{open} (one parameter) holds because for $w \in \mathbb{N}$ both values lie in $\mathbb{N}$ and for $w = a$ both lie in $\mathbb{N} \cup \{a\}$. But $\alpha + \beta = \alpha \neq \beta = \beta + \alpha$, so A_{xy} fails and, by soundness, is not derivable. With two parameters, $p_1 + p_2 = p_2 + p_1$ would be an instance of S_{ab}^{open} ; the convention excludes it. $\square$

Computed (`checks/check_models.out`): the 285 closed terms with at most 7 nodes denote naturals; the instances of S_{ab} , M_x , M_y hold for $x \in \{0, \dots, 14, \alpha, \beta\}$, and those of S_{ab}^{open} for 154 terms with at most 5 nodes and $w \in \{0, \dots, 3, \alpha, \beta\}$; A_{xy} and the instance $\forall x (x + p = p + x)$ of M_x^{open} fail at (α, β) .

A.4 Bounds, counters and further tables

Exact L1 pools. E1’s support check left out 65 (*open*) and 43 (*allq2*) data-derived theories for $\varphi = \neg Sx = 0$, summed over seeds, mostly skeleton clusters, and nothing else. *Dirichlet bounds* replaced the exact sum, with the largest posterior mass the theory could have had: in E1 for some skeleton theories and $\text{Mem}(D_n)$ under L1, at most $8 \cdot 10^{-14}$; in E2 for `skel6@32` and `skel6@64` in 14 cases over 25 seeds and both pools, 0 to double precision; in E6 for $\text{Mem}(D_n)$, 0; in E3 and E4(B, C), in no case checked. *Fallbacks.* The likelihood and derivability-oracle fallback counters were 0 in every run of E1, E3(a) and E4; the referee re-ran the 75 E1 jobs with the oracle instrumented (5896 queries, no fallback). *E4 queries.* $\forall x (x + 0 = x)$, $1 + 0 = 0$, $0 + 0 = 1$, $0 = S0$, $2 + 0 = 3$, $7 + 0 = 8$, $(1 + 1) + 0 = S(1 + 1)$, $0 = 0$; two are true. The code’s list has nine entries because it writes $1 + 0 = 0$ twice. *E5(a2).* Under $\{\varphi(z), \varphi(Sz)\}$ a datum $\varphi(t)$ has probability $Q(t)(1 - w_2 + w_2/q_S)$ if t is S -rooted and $Q(t)(1 - w_2)$ otherwise ($q_S = 0.35$), so $\text{BF} = \mathbb{E}[(1 - w_2 + w_2/q_S)^{n_S}(1 - w_2)^{n - n_S}]$, $w_2 \sim \text{Beta}(\frac{1}{2}, \frac{1}{2})$. Quadrature: substitution $w_2 = u^2$, $2 \cdot 10^5$ -point trapezoid rule; at $n = 500$ it agrees with the exact programme to 10^{-4} bits, and the referee’s exact Beta-moment sum agrees with it to $3 \cdot 10^{-4}$ bits up to $n = 10^6$.

n	Q seen	SeenQ(D_n)		frag-c.	CL(T) – CL(T^*), bits				Q-lumped	acc. seeds	
		mass	vs Q-l.		$+T_\wedge$	$+\forall x(0+x=x)$	$+0=S0$	legacy		causal	
8	2.5	0.744	−16.8	696.9	87.6	13.8	5.1	−100.0	18	2	
16	4.0	0.803	−38.1	698.2	87.9	14.2	5.4	−41.0	3	3	
32	5.7	0.923	−87.6	700.0	88.2	14.6	5.9	51.8	2	0	
64	6.7	0.991	−255.0	703.6	88.5	15.1	6.3	248.2	0	0	
512	7.0	0.996	−2928.5	716.3	89.2	16.6	7.8	2928.5	–	0	

Table 5: E2, 25 seeds. Left: the sound competitor $\text{SeenQ}(D_n) = \text{Trim}(T^*, D_n)$; its code length minus that of **Q-lumped** ranges over seeds from –49.1 to +8.9 at $n = 8$ and from –92.8 to +12.7 at 16. At $n \leq 32$ it is usually in the pool under the name of a data-derived theory; from $n = 64$ it is usually T^* or $T^* - Q_i$. Middle: code lengths relative to T^* (means; positive means worse); the over-general induction templates are 147 and 538 bits behind at $n = 8$, 8303 and 32624 at 512. Right: seeds in which $V_{0.05,0}$ accepts a false probe, with the first version’s legacy pool (data-derived theories from the first 16, 40, 64 data, used at every n ; no trimmed theories) and with the causal pool; at some $n \leq 64$, 19 against 5 seeds. Mean unsound mass, legacy against causal: 0.720/0.256, 0.147/0.197, 0.081/0.077, 0.009/0.009 at $n = 8, 16, 32, 64$. The legacy counts reproduce the referee’s. The referee’s variant, legacy plus SeenQ, accepts in seeds 1, 4, 11 ($n = 16$) and 17 ($n = 8$); the causal pool accepts in these and in seed 14 at $n = 8$, where $\text{Trim}(\text{Q-lumped})$ is the MAP with mass 1.000, so it is not uniformly kinder to soundness.

n	heavy data			numerals data			
	C_1	C_2	R_4	R_9	R_{11}	R_{12}	R_{16}
128	–0.2	–11.5	–162.0	–86.6	–3.0	+46.6	+300.2
512	–76.3	–204.4	–978.8	–1270.0	–1248.5	–1220.7	–1015.0
1024	–165.5	–447.0	–2063.1	–2835.1	–2894.6	–2895.1	–2751.7

Table 6: E3(a): code length relative to H_{sch} on the extended families (L0, mean over 5 seeds, bits; negative means preferred). The best R_m moves to larger m as n grows and is interior; C_2 pulls away from C_1 linearly. The first version’s pool stopped at R_4 and C_1 , which it named as the winners.

(A) tight construction								
u	w^*	δ'	t^*	exact $P(\text{win})$	MC (10^5)	pipeline	selected	bound
0.05	0.01	0.01	150	0.00046	0.00046	0/400	20/20	0.01
0.01	0.001	0.05	680	0.00108	0.00123	not run	20/20	0.05
0.5	0.01	0.01	10	0.00098	0.00095	2/400	20/20	0.01
0.1	0.1	0.1	47	0.00707	0.00694	5/400	20/20	0.1
0.2	0.05	0.2	17	0.02252	0.02298	6/400	20/20	0.2
(B, C) pools; L0 / L1; bound 0.05								
setting			prover wins		largest invalid mass		MAP at $n = 256$	
B well specified			0/100, 0/100		0.452, 0.533		$H_{\text{sch}} \times 100$	
C1 10% near misses $t + 0 = St$			100/100, 100/100		1.000, 1.000		$H_{\text{sch}} \cup \{z + 0 = Sz\} \times 100$	
C2 heavy-tailed terms			0/100, 0/100		0.452, 0.536		$C_1 \times 96, H_{\text{sch}} \times 4$	
C3 only $\varphi(0)$; decoy in pool			0/100, 0/100		0.107, 0.112		Mem $\times 100$	

Table 7: E4. (A) “Pipeline”: wins of the full posterior–derivability–threshold pipeline, which equals the closed-form event on all 400 streams in the two settings checked; “selected data”: only $\varphi(0)$ shown. (B) The largest invalid mass sits on $\forall x (x + 0 = x)$ at $n \leq 6$ ($n \leq 3$ in 184 of 200 streams), carried by H_{open} (mean 0.23 under L0, 0.21 under L1), by $H_{\forall}$ under L1 (0.15) and by a generalisation of $\varphi(z)$. (C1) First acceptance ($0 + 0 = 1$ or $2 + 0 = 3$) at the first mistaken datum in 90 of 100 streams per likelihood, 2–5 data later in the rest.

A.5 Reproduction and checks

All commands run from `code/` (Python 3.11, numpy 2.4, scipy 1.17). Results are in `results/*.md` and `/*.json`, the first version’s in `results/v1/`. AS’s `dtrc` code is read, not modified. `python3 -m pytest -q tests` runs the 21 unit tests (18s); `experiments/ek_*.py` runs `Ek`; `sh run_all.sh` runs everything (about 50 minutes on 4 shared cores; E3 10 min, E4 32 min, the rest at most 3 min each). Seeds: E1 0–4, E2 and E7 0–24, E3 and E6 0–4, E4 streams 0–399 (A) and 0–99 (B, C), E5 and E8 0–9; the random subsets for Min have seed $(c + \text{seed}) \cdot 1000 + b$ with $c = 1000$ (E1), 500 (E2), 77 (E3). After the batch run E1–E3 were re-run with their final reporting code; all posteriors are identical, and only derived labels changed. Two L0 entries of the first version’s E1 table were wrong on its own results and are corrected in Table 1. The unit tests compare each core computation with brute force or Monte Carlo, and with the referee’s independent `ref_core.py` (within 10^{-9}). Nothing is checked in a proof assistant. The check scripts (`research/tracks/experiments/checks/`, outputs next to them) give: `check_x6`, E1’s L1 odds within $1.1 \cdot 10^{-11}$ bits of prior $+n$ with prior differences (0.0899, 0.0899, 1.4404 bits) from the referee’s code, and the `L1citsel` tie to $1.0 \cdot 10^{-13}$; `check_fixed_weight`, ?? in this code (every $n \leq 2 \cdot 10^6$, 300 runs: 0.977, 1.000 at fixed w^* ; 0.003, 0.007 with w^* from the prior; 0 with the shrinking threshold) and $R_{\alpha}(n, 2) \geq 1/(e\pi(n + 1))$ for $n \leq 2000$; `kt_regret`, slopes $-0.499, -0.998, -1.485$ of $\log_2 R_{\alpha}(n, K)$ for $K = 2, 3, 4$; `check_models`, the structures of ?? and Proposition 1.2; `check_e4_tight` and `check_e4_where`, E4 as in Table 7; `check_e1_v1`, E1’s generator masses unchanged against the first version to three decimals; `check_bounded` and `check_classprob_open`, as in Appendices A.1 and A.4. The referee’s own code reproduced the first version’s E1–E7 bit for bit, and the final code reproduces the referee’s legacy E2 counts, E3 extensions and atomic-motive run.

spare added to $\{\varphi(z)\}$	mean $\log_2$ BF at $n = 1, 16, 256, 4096$	slope	prior bits
false sentence $0 = S0$ (never cited)	-1.00, -2.84, -4.83, -6.83	-0.500	8.4
disjoint schema $z \cdot 0 = 0$ (never matched)	-1.00, -2.84, -4.83, -6.83	-0.500	14.5
nested $\varphi(Sz)$	+0.17, -1.05, -2.85, -3.97	-0.281	21.3
nested $\varphi(Sz)$, quadrature	$n = 10^2 \dots 10^8$: see caption	-0.251 ± 0.002	21.3

Table 8: E5 (10 seeds; Bayes factors without the prior; slopes in $\log_2$ BF per doubling of n , on $n = 256 \dots 4096$ except for the quadrature row, $10^2 \dots 10^8$). (a2) The nested spare by quadrature: mean $\log_2$ BF $-2.196, -3.028, -3.867, -4.658, -5.575, -6.332, -7.200$ at $n = 10^2, \dots, 10^8$ (s.e. $0.028\text{--}0.031$); slope -0.251 ± 0.002 on $10^2 \dots 10^8$ and -0.251 ± 0.003 on $10^4 \dots 10^8$; per decade $-0.250, -0.253, -0.238, -0.276, -0.228, -0.261$; 1000 draws of $n_S \sim \text{Bin}(n, 0.35)$ per n , numpy seed 0; the first version reported -0.259 from 200 draws, and the referee’s independent sum -0.233 on $10^2 \dots 10^7$ with 40 draws. (b) L_5 data: L_5 has mass $2 \cdot 10^{-6}$ at $n = 64$, 0.66 at 128, 1.000 from 256 (10/10 seeds); L_∞ data: L_∞ has 1.000 from $n = 16$. $\mathcal{Q}_{\text{num}}(S^j 0) = 2^{-(j+1)}$; prior bits L_1 10.9, L_5 89.4, L_8 187.9, L_∞ 17.1; per datum L_5 gains $3 - \log_2 5 \approx 0.68$ bits, which repays the 72-bit prior difference and L_5 ’s Dirichlet cost of about $2 \log_2 n$ bits between $n = 64$ and 128. (c) Stage lengths 1, 24, 81, 115, 104, 83, 68, 55, 45, 39, 33, 33, 32, 27 (740 data); they shorten after $i = 4$ because L_∞ ’s mean cost on round-robin data, $1 + (i - 1)/2$ bits, outgrows L_i ’s $\log_2 i$.

data from	generator’s mass at $n = 2, 4, 8, 16, 256$	first n (≥ 0.99), per seed	$P(\text{proves } A_{xy})$ at $n = 4; 256$
A_{xy}	0.34, 0.90, 1.000, 1.000, 1.000	8, 8, 8, 4, 2	1.000; 1.000
M_x	0.60, 0.80, 0.90, 0.99, 1.000	32, 16, 8, 16, 16	0.205; $7 \cdot 10^{-49}$
S_{ab}	0.28, 0.56, 0.95, 1.000, 1.000	16 in every seed	0.060; $1 \cdot 10^{-11}$

Table 9: E6 under L1 (means over 5 seeds). Under $L1_{\text{cit}}^{\text{sel}}$ at $n = 256$, for all three sources: A_{xy} 0.044, A_{yx} 0.044, S_{ab} 0.128, M_x 0.392, M_y 0.392, open forms $\leq 6 \cdot 10^{-9}$, so $P(\text{proves } A_{xy}) = 0.088$. Prior bits: A_{xy}, A_{yx} 31.6; $S_{ab}, S_{ab}^{\text{open}}$ 30.0; the four M forms 28.4. Provability of A_{xy} is read from Proposition 1.2, not from $\vdash_J$.

References

- Claude (Anthropic). Learning axioms and axiom schemas from their instances, 2026a. Draft report prepared for Kaarel Hänni; **axiom-schemas/** in the accompanying repository.
- Claude (Anthropic). What follows from what: Learning inference rules from imitation, coherence, and the world. Unpublished report prepared for K. Hänni, 2026b. **inferential-learning/** in the accompanying repository.

n	<i>wellspec</i> , L1	<i>wellspec</i> , L0	<i>skew</i> , L1	<i>skew</i> , L0
16	-1.7 (-3.4, 0.8)	-2.2 (-4.5, 1.2)	1.7 (-1.5, 8.3)	-0.2 (-2.3, 5.6)
64	-3.5 (-5.5, 1.2)	-2.8 (-5.2, 0.1)	7.7 (-0.4, 14.3)	1.2 (-2.9, 5.7)
256	-7.3 (-8.3, -6.1)	-6.8 (-7.8, -5.6)	48.8 (37.9, 72.6)	17.4 (6.4, 34.2)
1024	-9.8 (-11.6, -6.5)	-9.3 (-11.4, -6.2)	196.5 (165.4, 232.8)	72.1 (41.9, 98.0)
4096	-13.0 (-14.6, -11.3)	-12.9 (-14.5, -11.4)	817.0 (738.9, 912.3)	299.6 (210.7, 351.5)

Table 10: E8: $\log_2$ Bayes factor split : whole without the prior (mean over 10 seeds; range).

(λ, τ)	(1, 0)	(1, 1)	(1, 4)	(2, 0)	$(\frac{1}{2}, 0)$
T^* at $n = 32$	0.197	0.197	0.197	0.040	0.186
T^* at $n = 64$	0.751	0.751	0.752	0.594	0.719
T^* at $n = 128$	0.991	0.992	0.993	1.000	0.961
T^* at $n = 512$	0.996	0.996	0.996	1.000	0.980
unsound at $n = 8$	0.256	0.293	0.372	0.860	$3 \cdot 10^{-4}$
unsound at $n = 16$	0.197	0.217	0.262	0.792	$7 \cdot 10^{-5}$
unsound at $n = 32$	0.077	0.084	0.103	0.737	0.014
unsound at $n = 64$	0.009	0.009	0.008	0.206	0.039
unsound at $n = 512$	0.004	0.004	0.004	$2 \cdot 10^{-4}$	0.019
acc. at $n = 8, 16, 32, 64$	2, 3, 0, 0	2, 3, 0, 0	5, 4, 1, 0	18, 19, 18, 5	0, 0, 0, 0
MAP at $n = 8$: e/w/u	0/18/7	0/16/9	0/16/9	0/3/22	0/25/0

Table 11: E7 (E2’s data and pools; means over 25 seeds). “Acc.”: seeds in which $V_{0.05,0}$ accepts a false probe; no seed accepts from $n = 128$. Last column: MAP tags equivalent / weaker / unsound at $n = 8$. T^* has mass below 10^{-3} at $n \leq 16$ in every row. $\lambda = \frac{1}{2}$ is proper only on finite pools. The first version (5 seeds, legacy pool) reported a largest time-factor effect of 0.002 and 0.016; the referee’s re-run of that pool with SeenQ gave an unsound MAP at $n = 8$ in 19 of 25 seeds at $\lambda = 2$, the same direction as here.